

CLASS 08 / OCT 13 / U05

ATTRACTORS

2D FIELD DRAWING

P3 CONTINUES — A FIELD FIRST, THEN A DRAWING MADE OF LINES

P3b DUE

P3c ISSUED TODAY

AN ATTRACTOR IS A NUMBER

For every point: how far is it from something? Use that number to drive a property.

THE WHOLE IDEA

```
for each point p:  
    d = distance(p, a)  
    value = f(d)
```

Everything else this week
is a choice of f.

MEASURE IT FIRST

$$d = \sqrt{(px - ax)^2 + (py - ay)^2}$$

Pythagoras, once per point. The answer is in model units.

Index told you where in the sequence. Distance tells you how far from something outside the collection.

LAST WEEK YOU BUILT THE POSITIONS. THIS WEEK YOU MEASURE THEM.

READ THE FIELD BEFORE YOU CHANGE ANYTHING

Distance is a value at every point. Look at it as grey before it becomes geometry.

01 MEASURE

Every point on a flat 2D grid gets a number: how far to the attractor.

02 DISPLAY

Show those numbers as grey.
This is a drawing already.

03 REMAP

Push that range into the range your property needs.

04 DRIVE

Only now does anything change. The field was finished first.

THE FIELD, READ AS A GREY VALUE ACROSS THE PLANE · NEAR → FAR



IF THE GREY IMAGE IS NOT ALREADY A GOOD DRAWING, THE LINES WILL NOT SAVE IT.

P3c requires the grayscale field, with a legend, beside the line drawing. That order is the assignment.

REMAP, AND SAY WHERE YOU CLAMPED

Distance comes out in model units. Your property needs a specific range.

THE OPERATION

```
t    = (d - dmin) / (dmax - dmin)
t    = clamp(t, 0, 1)
out  = outmin + t × (outmax - outmin)

d    0 ... 8.4 units    ← input
out  2 ... 12 mm length ← output
```

SUBTRACT. DIVIDE. MULTIPLY. ADD.

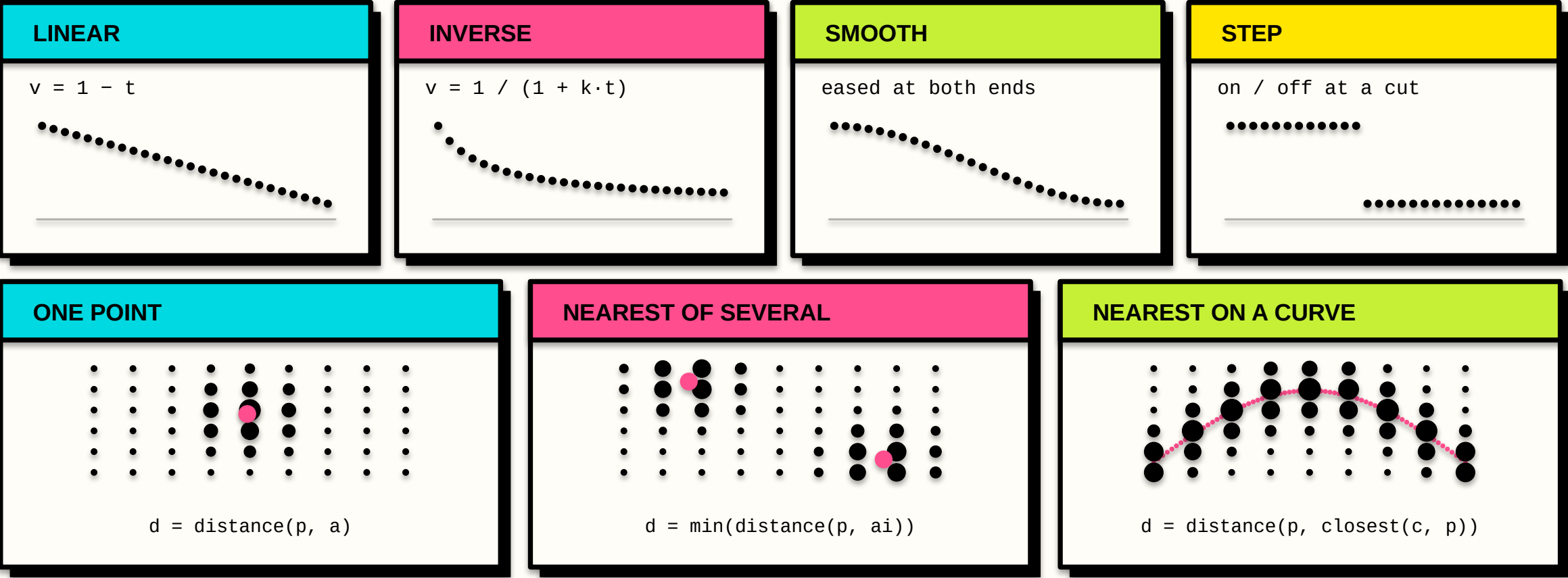
Nothing here is new. It is the arithmetic of a transformation, applied to a distance instead of a coordinate.

Clamping is a decision, not a safety net. Decide what happens past the end of your range and say so.

P3c REQUIRES INPUT AND OUTPUT RANGES, CLAMPING AND FALLOFF LABELLED ON THE SHEET.

FALLOFF, AND WHAT YOU MEASURE TO

f shapes the drawing. The attractor type changes only what the distance is measured to.



SAME GRID, SAME MAPPING, THREE ATTRACTORS: P3c SHOWS WHAT THE ATTRACTOR ALONE CHANGES.

A LINE DRAWING IS LINES, NOT PIXELS

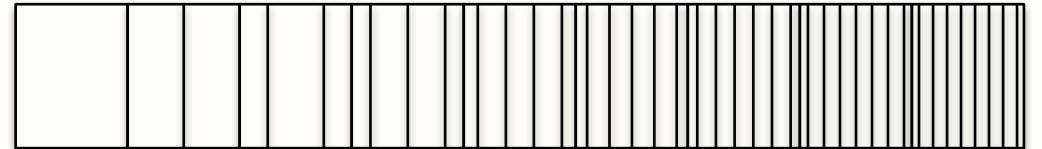
A screen can show any grey. A line, drawn or plotted, is either there or not.

ON SCREEN — A VALUE PER PIXEL



Every sample can be any grey. Nothing needs to be drawn.

AS LINES — A STROKE OR NOTHING



Every stroke is the same darkness. Grey comes from how many lines, how long and how close.

SO THE FIELD MUST BECOME GEOMETRY: CURVES WITH A START AND AN END. THAT IS WHAT THE SVG HOLDS.

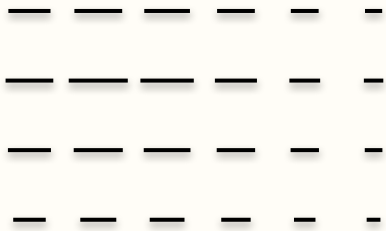
The grayscale field stays on the sheet, with its legend, beside the line drawing. The lines are what you submit as SVG — and what a pen plotter could draw.

TRANSLATE THE FIELD INTO LINES

Same field, same remap. Only what the output number drives changes. Pick one.

LINE LENGTH

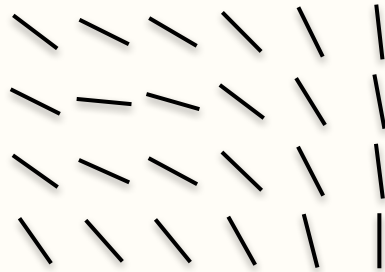
length = out



Long near, short far.

ROTATION

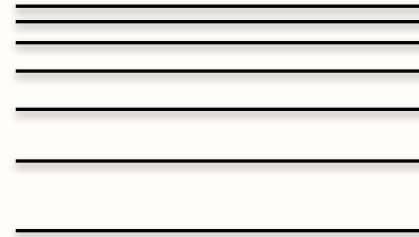
angle = out



Lines turn with distance.

SPACING

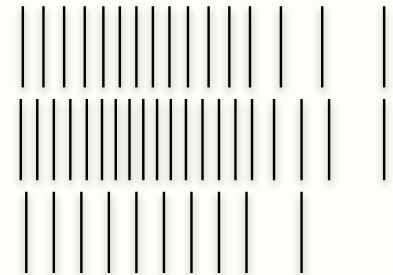
gap = out



Rows open up with distance.

HATCH DENSITY

lines per cell = out



More strokes, darker cell.

```
t = remap(distance(p, a))      # the field, unchanged
line(p, angle, length)         # the last line is the translation
```

EXPORT THE SVG AT SIZE. PLOTTING IS OPTIONAL.

The SVG is required. If you choose to plot it, test small before the pen touches paper.

01 EXPORT THE SVG

CURVES

Keep strokes as curves. No mesh, no filled shape.

SIZE

17 × 11 in, the sheet size, at 100%.

UNITS

Know the file unit. Measure one known line.

PEN

One pen weight, or one layer per declared pen.

MARGIN

Every stroke inside the paper, with a margin.

02 OPTIONAL — TEST PLOT SMALL FIRST

CROP

A small, dense corner of the real drawing.

PEN

Check tip and ink. A dry pen skips.

PAPER

Real paper, fixed flat, right way round.

SPEED

Slower for dense areas. Watch the first strokes.

COMPARE

Test beside screen. Change the mapping, not the pen.

03 COMMON PROBLEMS — EACH ONE IS VISIBLE IN THE SVG BEFORE IT IS PRINTED OR PLOTTED

DUPLICATE LINES

Strokes on top of each other. Remove overlaps.

TOO-DENSE HATCHING

Lines closer than the pen blot. Space them wider.

FILLS

A fill has no stroke to follow. Turn it into hatching.

TINY STROKES

Thousands of short segments. Simplify or lower the count.

ISSUED TODAY

P3c — ATTRACTOR: 2D FIELD DRAWING

DUE WEEK 10 — Oct 27 / 7%

Sample a field on a flat grid, turn every value into a line, and present the line drawing beside its grayscale field on one sheet. Same grid, same mapping, three attractors: the drawing shows what the attractor alone changes.

FIELD

GREY FIRST

The grayscale field, with a legend, before any line.

LINES

ONE MAPPING

Length, rotation, spacing or hatch density, driven by distance.

SHEET

ONE 17 × 11

Line drawing, field and legend. SVG submitted. Plotting is optional.

REQUIREMENTS — P3c

Visual quality 35% · computational understanding 25% · technical execution 20% · experimentation 10% · requirements and identity 10%.

01

Build a 2D attractor field on a flat grid and translate it into lines — vary line length, rotation, spacing or hatch density with the measured distance.

02

Compare single-point, multiple-point and curve attractors on the same grid, with the same mapping, so the difference comes from the attractor alone.

03

Make one 17 × 11 inch sheet presenting the line drawing, with the grayscale field and a legend beside it. Label input and output ranges, clamping and falloff.

04

Keep the drawing as clean vector lines and submit the SVG with the sheet PDF and the editable graph.

05

Optional — plot the drawing on the pen plotter and add a photograph or scan of the plot to the sheet.

CHECKPOINT · WEEK 9 — Oct 20 — Bring the grayscale field and a first line drawing of it.

NOW — FIELD TO DRAWING

Rest of the class: build the field, translate it, export it, start the sheet.

✓

Build the 2D field: a flat grid, distance at every sample, shown as grey with a legend

✓

Write down input and output ranges, clamp and falloff — actual numbers

✓

Translate the field into lines: length, rotation, spacing or hatch density

✓

Export an SVG at sheet size and measure one known line

✓

Optional: run a small test plot on the pen plotter and compare it with the screen

MOVE THE ATTRACTOR. IF YOU CANNOT PREDICT WHAT HAPPENS, STOP AND READ THE FIELD.

BEFORE NEXT CLASS

NEXT

Week 9 — Oct 20: layered site analysis — map a site's point cloud in layers with attractor logic. P3d is issued.

P3c

Checkpoint next class. Bring the grayscale field and a first line drawing of it.

DEADLINE

Week 10 — Oct 27 is the mid-semester deadline. GSM, P2b and P3c are due, and everything through Arrays must be in.

SITE

Think of a site you can document. P3d starts from its point cloud.